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The SET-Domain-Containing Protein MdSDG26 Negatively Regulates *Alternaria alternata* Resistance in Apple

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ABSTRACT

Apple leaf spot is one of the most devastating diseases in the apple industry, caused by *Alternaria alternata* f. sp. *mali* (*A. alternata*). SET-domain group (SDG) proteins function as the histone methyltransferases and participate in plant development and stress responses. However, whether SDG proteins are associated with *A. alternata* resistance is largely unclear. Here, we describe the pathogen-inducible *MdSDG26* gene in apple (*Malus × domestica*). *MdSDG26* has two transcript variants that function similarly in catalyzing histone methylation and *A. alternata* resistance. Transient overexpression of *MdSDG26* increased the global levels of H3K4me3 and H3K36me3, whereas knockdown of *MdSDG26* only reduced the H3K36me3 level. Transcriptome analysis revealed that *MdSDG26* affected the genome-wide transcriptome changes in response to *A. alternata* infection. ChIP-qPCR analysis demonstrated that *MdSDG26* modulates the levels of H3K36me3 and H3K4me3 at both the promoter and exon regions of *MdNTL9*. As a negative regulator of *A. alternata* resistance in apples, *MdNTL9* plays a pivotal role in *MdSDG26*-mediated resistance to *A. alternata*. Therefore, our findings provide compelling evidence for the regulatory function of *MdSDG26* in histone methylation and its molecular role in conferring resistance to *A. alternata*.

1 | Introduction

Apple is one of the most important fruit crops in the world, but cultivation is always affected by biotic stresses, especially pathogens. Apple leaf spot is a devastating fungal disease caused by the fungal pathogen *Alternaria alternata* f. sp. *mali* (*A. alternata*), which often reduces apple production (Moriya et al. 2019). The Resistance (R) proteins containing the

CC_R-NB-LRR (coiled-coil, nucleotide-binding and leucine-rich repeat) domain confer resistance to *A. alternata* in apples. Five R proteins, MdRNL1, MdRNL2, MdRNL3, MdRNL4 and MdRNL5, positively regulate *A. alternata* resistance in apple, and hairpin RNA (*hpRNA*) *MdhpRNA277* derive small interfering RNA (siRNA), *siR277s*, which are induced by *A. alternata*, can target these five R protein genes and negatively regulate *A. alternata* resistance (Zhang et al. 2018). MdRNL2

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and MdRNL6 form a complex and interact with pathogenesis-related (PR) proteins, MdPR10-1 and MdPR10-2, that positively regulate the *A. alternata* resistance (Zhang, Xu, et al. 2021). MdMYB88 and MdMYB124 positively regulate *A. alternata* resistance through the phenylpropanoid biosynthesis pathway (Geng et al. 2020). MdHYL1 (double-stranded RNA-binding protein), the interacting factor of MdMYB88 and MdMYB124, positively regulates *A. alternata* resistance by mediating expression of *MdMYB88* and *MdMYB124* and biogenesis of several microRNAs responsive to *A. alternata* infection, such as mdm-miR160, a negative regulator to *A. alternata* resistance (Shen et al. 2023).

During the co-evolution with host-adapted pathogens, plants have evolved complex systems to defend against pathogen attacks (van der Burgh and Joosten 2019; Zhou and Zhang 2020). Pattern-triggered immunity (PTI) and effector-triggered immunity (ETI) are mutually potentiated in plant immunity, and both of them are associated with a massive transcriptional reprogramming of defense-related genes (Birkenbihl et al. 2017; Adachi and Tsuda 2019; Bjornson et al. 2021; Ngou et al. 2021; Pruitt et al. 2021; Yuan et al. 2021; Zhi and Chang 2021). Transcription factors (TFs) are involved in the ETI pathway by regulating the transcription and expression of genes essential for defense responses, such as NAC (No apical meristem [NAM], Arabidopsis transcription activation factor [ATAF], Cup-shaped cotyledon [CUC]) superfamily of TFs (Puranik et al. 2012; Guo et al. 2017; Johnsson et al. 2019; Li, Zeng, et al. 2021). NAC TRANSCRIPTION FACTOR-LIKE 9 (NTL9) plays a negative role in regulating disease resistance in Arabidopsis (Guo et al. 2017; Wang et al. 2019). NTL9 can interact with the *Pseudomonas syringae* type III effector HopD1, which suppresses effector-triggered immunity and leads to the inhibition of immune marker gene transcription (Marin-Ponce et al. 2023). NAC81, also named ATAF2, can be induced by ethyl jasmonate, salicylic acid and pathogen infection (Hauck et al. 2003; Delessert et al. 2004). Plants overexpressing ATAF2 display higher susceptibility to the soil-borne fungal pathogen *Fusarium oxysporum*, underscoring the regulatory role of ATAF2 in plant defense mechanisms against pathogens (Delessert et al. 2005).

Increased evidence has indicated that the epigenetic process plays key roles in the regulation of plant disease resistance, such as histone methylation (Zhang, Hung, et al. 2021; Zhi and Chang 2021; Kang et al. 2022). Histone lysine residues on H3 can undergo mono-, di-, or tri-methylated at positions K4, K9, K27 and K36, facilitated by histone methyltransferases (Liu et al. 2010). Plant histone methyltransferases possess a conserved SET [Su(var)3-9, Enhancer of Zeste E(z) and Trithorax] domain that is essential for their enzymatic activities (Thorstensen et al. 2011). The SET-domain group (SDG) protein, a class of histone methyltransferases in plants (Li, Yan, et al. 2021), can be categorized into seven classes based on sequence homology and phylogenetic relationships (Ng et al. 2007; Zhou et al. 2020; Li, Yan, et al. 2021). Besides the SET domain, the adjacent domains in SDG proteins can recognize modules in chromatin or act as binding platforms for their interaction partners. Thus, different SDG classes are responsible for catalyzing the different lysine methylation sites (Schapira 2011; Zhou et al. 2020). Generally, H3K9 or H3K27 methylation is regulated by E(z), Suv, and ATXR5/6 proteins for gene

repression, whereas H3K4 or H3K36 methylation is catalyzed by Ash and Trx proteins for gene activation (Ng et al. 2007; Li, Yan, et al. 2021). Histone methylation plays an important role in plant development and biotic and abiotic stress resistance (Yuan et al. 2013; He et al. 2021; Kang et al. 2022).

Previous studies have shown that SDG proteins are involved in plant disease resistance. In Arabidopsis, AtSDG8 plays a role in the catalysis of H3K36me2/3 and is necessary for the deposition of H3K36me3 at *LAZ5* that encodes a pathogenesis R protein (Xu et al. 2008; Palma et al. 2010). AtSDG8 also plays a vital role in disease resistance by modulating a range of genes in jasmonic acid and/or ethylene signalling pathways (Berr et al. 2010). In tomatoes, *SISDG33* and *SISDG34* (the homology gene of AtSDG8) are required for pathogen-induced enrichment of H3K36me3 and H3K4me3 at the target genes. Loss of function of *SISDG33* and *SISDG34* increased plant disease resistance (Bvindi et al. 2022). In Arabidopsis, SDG8 could interplay with SDG26 to regulate transcription and plant flowering and development, the early-flowering phenotype of the *sdg8* mutant is epistatic to the late-flowering *sdg26* mutant, *sdg8 sdg26* double mutant exhibiting defects similar to *sdg8* mutant but opposite to *sdg26* mutant (Liu et al. 2016). In the *sdg26* mutant Arabidopsis plants, global levels of H3K36me1, H3K36me3 and H3K4me3 were comparable to those in wild-type plants. However, these histone methylation marks may exhibit locus-specific alterations that contribute to plant development (Xu et al. 2008; Liu et al. 2016). The function of SDG26 in disease resistance is still not well understood.

However, the involvement of histone methylation, particularly that catalyzed by SDG protein, in *A. alternata* resistance in apples remains largely unexplored. Here, we found that overexpression of *MdSDG26* enhanced the global levels of H3K4me3 and H3K36me3, whereas knockdown of *MdSDG26* specifically reduced global H3K36me3 levels. Additionally, *MdSDG26* was found to regulate the expression of disease-related genes, thus playing a negative role in regulating *A. alternata* resistance in apple.

2 | Materials and Methods

2.1 | Plant Materials and Growth Conditions

Tissue-cultured GL-3 (the progeny of the 'Royal Gala' apple, which was used as the background for apple transformation) and transgenic plants were subcultured every 4 weeks on the Murashige & Skoog (MS) medium and grown under long-day conditions at 25°C, as previously described (Niu et al. 2022). After rooting, GL-3 and transgenic plants were transplanted into soil and grown for an additional 3 months in a growth chamber at Northwest A&F University, Yangling (34°20'N, 108°24'0'E), China.

2.2 | RNA Isolation and RACE Experiment

The CTAB method was used for RNA extraction from GL-3 leaves, and the genomic DNA was removed by DNase I

(Thermo Scientific, EN0521, USA) treatment as described previously (Chang et al. 1993; Xie et al. 2018). Single-stranded cDNA was synthesized from the high-quality total RNA with RevertAid reverse transcriptase and a specific oligo-dT₁₈ adaptor using the RevertAid First Strand cDNA Synthesis Kit (Thermo Scientific, K1622, USA). Based on the predicted gene sequence (Daccord et al. 2017), gene-specific primers (GSP) were designed for Rapid-Amplification of 3'-cDNA Ends (3'RACE), Rapid-Amplification of 5'-cDNA Ends (5'RACE) or gene cloning. For 3'RACE, GSP and 3'RACE Out-primer were used for the first round of PCR amplification based on the cDNA template, then the PCR products were used as the templates for the second round of PCR amplification with GSP and 3'RACE inner-primer. Finally, the PCR products were recovered from an agarose gel and then introduced into a T-vector (Yeasen, 10907ES20, China) for sequencing. For 5'RACE, the cDNA was amplified using a suitable 5'RACE protocol provided in the HiScript-TS 5'/3'RACE Kit (Vazyme, RA101-01, China). The 3'/5'RACE primers were provided by the kits and the GSPs' positions are marked in Figure 1C. The PCR products were

recovered from an agarose gel and then introduced into a T-vector (Yeasen, 10907ES20, China) for sequencing. The primers used are listed in Table S1.

2.3 | Vector Construction and Genetic Transformation

Using the Gateway technology, the ORF (open reading frame) sequence of *MdSDG26.1*, *MdSDG26.2* and *MdNLT9* was introduced into the binary vector pEarleyGate101 to generate the overexpression vector. A 358-bp fragment of *MdSDG26* was introduced into the binary vector pK7GWIWG2D to generate the RNAi vector. The plasmid of *MdSDG26* RNAi was transformed into *Agrobacterium tumefaciens* strain EHA105, followed by transformation into GL-3 as previously described (Dai et al. 2013; Niu et al. 2022) to obtain *MdSDG26* RNAi transgenic apple plants. The plasmids for *MdSDG26.1*, *MdSDG26.2* and *MdNLT9* overexpression were transformed into *A. tumefaciens* strain EHA105, followed by transient expression in 3-week-old

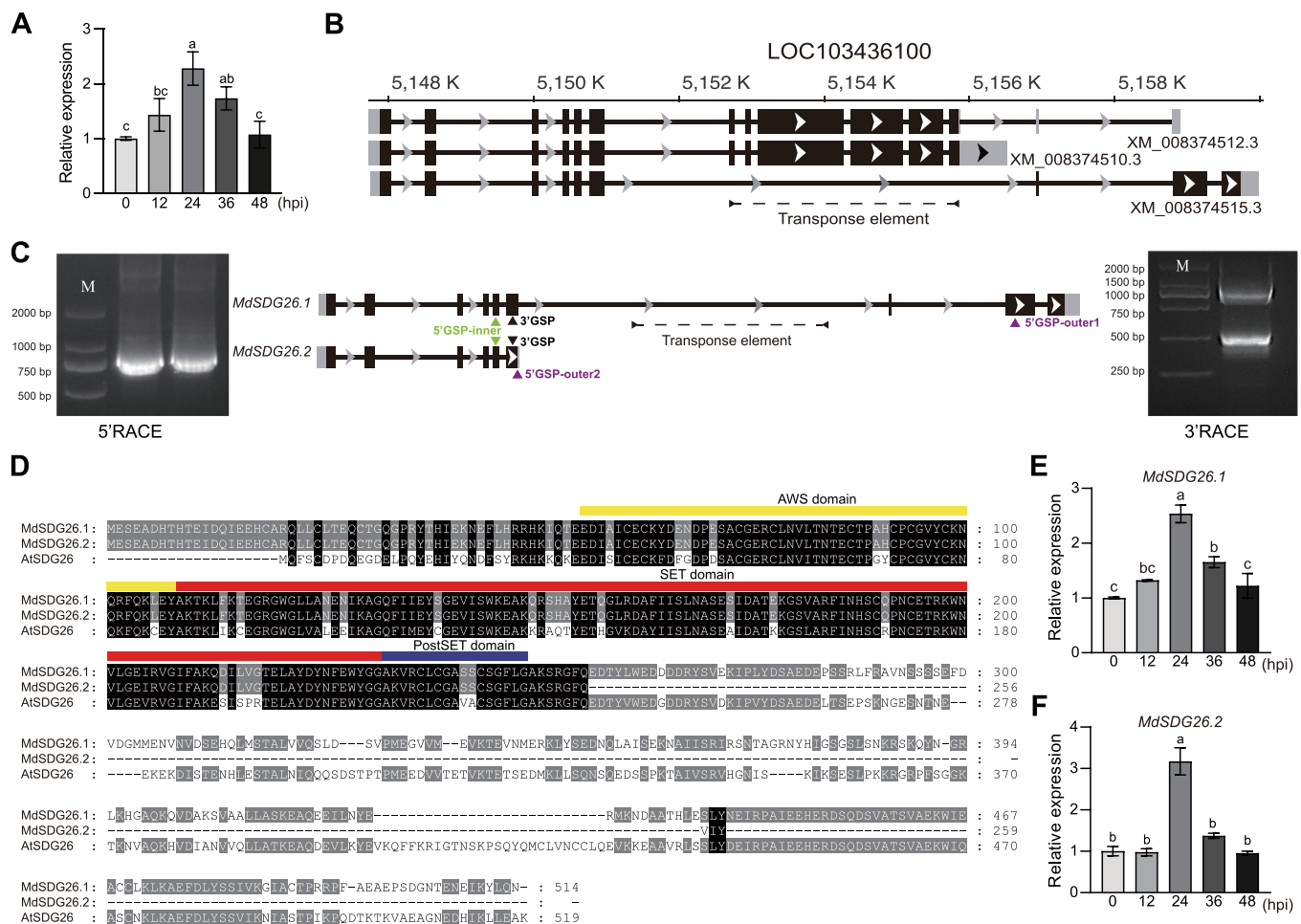


FIGURE 1 | Pathogen response gene *MdSDG26* has two transcript variants in apple. A, Expression of *MdSDG26* in leaves of GL-3 under Mock and *Alternaria alternata* treatment. hpi, hours postinfection. B, The predicted transcript variants of *MdSDG26* by the NCBI website. Exons and introns were indicated by boxes and lines; the grey box indicated Untranslated Regions (UTR), and the gene transcript direction was marked as arrows. C, 3'RACE and 5'RACE results of *MdSDG26* and the diagrams of its transcript variants. M, Marker. GSP, gene-specific primer. D, Protein sequence alignment of *MdSDG26.1*, *MdSDG26.2* and *AtSDG26*. At, *Arabidopsis thaliana*. Conserved domains were indicated by yellow, red and blue squares. E, F, Expression of *MdSDG26.1* (E) and *MdSDG26.2* (F) in leaves of GL-3 under Mock and *A. alternata* treatment. Error bars indicate SD, $n = 3$. Statistical analyses were performed by one-way ANOVA (Tukey's test). The letters indicate significant differences ($p < 0.05$).

GL-3 or *MdSDG26* RNAi plant plantlets (~ 70 leaves) as previously described (Liu et al. 2021). After 3 days, nine leaves from three different plantlets were sampled as one replicate, and three replicates were utilized for gene expression analysis. The experiments were performed twice.

2.4 | Subcellular Localization

Subcellular localization was performed as described by Xie et al. (2018). The ORF regions of *MdSDG26.1* and *MdSDG26.2* were cloned into pGWB406 vector and then transformed into *Agrobacterium* strain C58C1, followed by co-infiltration with 35S: *P19* and *NLS-mCherry* (a plasmid used for the nuclear localization signal, Addgene accession number: 108881, (Ariotti et al. 2018)) into 4-week-old tobacco (*Nicotiana benthamiana*) leaves. After 3 days, the infiltrated tobacco leaves were used for fluorescence signal observation. The experiments were performed twice.

2.5 | Fungal Growth and Disease Resistance Analysis

A. alternata f. sp. *mali* was cultured on Potato Dextrose Agar (PDA) medium at 25°C for about 6 days. The infection of *A. alternata* was carried out as previously described (Geng et al. 2020) with minor modifications. Briefly, more than 40 leaves of transiently overexpressed apple plants or the *MdSDG26* silenced apple plants were punctured by a needle and inoculated with a 1 cm fungi plaque or empty plaque (as the Mock). After inoculation, the leaves were returned to the culture dishes and placed in an incubator. The pictures were taken after 5 days for the transiently overexpressed plants and ~2 weeks for the *MdSDG26* RNAi plants, followed by measurement of the leaf spots by ImageJ software (Schneider et al. 2012). The experiments were performed at least twice.

2.6 | RNA-Seq Analysis and RT-qPCR Analysis

The apple leaves obtained from GL-3 and *MdSDG26* RNAi transgenic plants under Mock and *A. alternata* infection conditions were used for RNA extraction. Thirty leaves were used for the experiment, and ten leaves were used as one replicate. The RNAs were sequenced using the Illumina HiSeq platform by Novogene (Beijing, China), and then the reads were aligned with the GDDH13 reference genome version 1.1, followed by RNA-seq analysis as previously described (He et al. 2022). Gene Ontology (GO) analysis was performed on the website of agriGO v2.0 (Tian et al. 2017), and Gene Regulatory Network (GRN) was constructed from the different expression genes by Arabidopsis Interactions Viewer (http://bar.utoronto.ca/interactions/cgi-bin/arabidopsis_interactions_viewer.cgi). RT-qPCR analysis was carried out as described previously (Niu et al. 2022). For detecting the expression of *MdSDG26.1* and *MdSDG26.2*, a specific reverse primer based on the nucleotide difference and a common forward primer were used for RT-qPCR. Three biological replicates with three technical repeats were used. The primers used for RT-qPCR are listed in Supplemental Table 1.

2.7 | Histone Methylation Analysis

To detect the histone methylation status in vivo, total proteins were extracted from fresh leaf tissues as described by Li et al. 2023. Briefly, leaf tissues of 1-month-old tissue-cultured GL-3 or transgenic plants were ground into a fine powder in liquid nitrogen, then suspended in a 3× loading buffer [250 mM Tris-HCl (pH = 6.8), 10% sodium dodecyl sulfate (w/v), 0.5% bromophenol blue (w/v), 50% glycerol (v/v), 5% β-mercaptoethanol (v/v)] and boiled for 8 min, followed by centrifugation at 15 000g before the supernatant was loaded on the SDS-PAGE gel. Western blot was performed using the anti-H3 (ABclonal, A2348, China), anti-trimethyl-H3K4 (ABclonal, A2357, China) and anti-trimethyl-H3K36 antibodies (ABclonal, A2366, China), and the immunoblot signal with anti-H3 was used as an input control. The intensity of individual bands was quantified using ImageJ software (Schneider et al. 2012). Three biological replicates were applied and 12 plants were used for the experiment, with pooled leaves from 4 plants as one biological replicate.

2.8 | Chromatin Immunoprecipitation (ChIP) Assay

ChIP assays were performed according to the previously described method (Xie et al. 2018; Bvindi et al. 2022). Briefly, 4-month-old GL-3 and *MdSDG26* RNAi transgenic plants, which were treated by Mock inoculation and *A. alternata* inoculation after 24 h, were used for cross-linking. The chromatin was extracted and sonicated to shear DNA into 0.3–0.7 kb fragments, followed by centrifugation at 15 000g for 5 min at 4°C and transfer the supernatant into new tubes, and the supernatant was precleared with ChIP-grade protein A/G magnetic beads (Thermo Scientific, 26162, USA). Then immunoprecipitation was performed with or without antibodies against anti-trimethyl-H3K4 (A2357, ABclonal, China) and trimethyl-H3K36 (ABclonal, A2366, China). The chromatin antibody complex was precipitated with ChIP-grade protein A/G magnetic beads and washed with different washing buffers. Finally, DNA was recovered by a reverse crosslink and protein removal by protein K (Thermo Scientific, 25530049, USA). The recovered DNA was used for ChIP-qPCR analysis. The ChIP-qPCR primers used were designed in the promoter and coding region of genes. Rabbit immunoglobulin G (IgG) served as a negative control for immunoprecipitation of all chromatin samples. The enriched signal was normalized to the input signal and presented as a percentage of input (Input %). All the leaves collected from nine plants were pooled as one biological replicate. Three biological replicates were used for the experiment, and the experiment was repeated twice. The primers used for ChIP-qPCR are listed in Table S1.

2.9 | Statistical Analysis

Data were reported as the mean ± SD. Statistical significance was determined by Student's *t* test or one-way ANOVA (Tukey's test) analysis using SPSS (version 22.0, IBM, USA). Differences were considered significant if *p* < 0.05, 0.01, or 0.001.

3 | Results

3.1 | Disease-Induced *MdSDG26* Has Two Transcript Variants

Previously, we showed that *MD05G1031300* is induced by *Venturia inaequalis* infection in apple leaves, which encodes a class II SDG histone methyltransferases (Li, Yan, et al. 2021). This gene is homologous to SDG26 in Arabidopsis and we therefore named it *MdSDG26*. RT-qPCR results demonstrated that *MdSDG26* could be induced by *A. alternata* infection, and its highest expression level was detected at 24 h postinoculation (hpi) (Figure 1A). To investigate the function of *MdSDG26* and to clone the full-length sequence of *MdSDG26*, we analysed all the transcripts based on the NCBI (<https://www.ncbi.nlm.nih.gov/>). A long insertion was found after the sixth exon of *MdSDG26*, and then we performed analysis by CENSOR software (<https://www.girinst.org/censor/index.php>, Kohany et al. 2006) using the ~4000-bp sequence as a query and found that it was most similar to MuDR-4, a member of the MuDR-type DNA transposons. NCBI showed that *MdSDG26* (LOC103436100) had three transcript variants, two variants (XM_008374512.3 and XM_008374510.3) contained sequences transcribed from a transposable element, whereas one variant (XM_008374515.3) did not (Figure 1B).

To verify this prediction, we performed 3'RACE and 5'RACE experiments. The results showed that only two transcripts existed in the apple, one had the same sequence as XM_008374515.3 and another one was a new transcript, we named them *MdSDG26.1* (XM_008374515.3) and *MdSDG26.2*, respectively (Figure 1C). 5'RACE showed *MdSDG26.1* and *MdSDG26.2* had same 5'UTR of mRNA (Figures 1C and S1). Protein sequence alignment showed that *MdSDG26.1* and *MdSDG26.2* retained the conserved domains with AtSDG26 and AtSDG8 (Figures 1D and S2), including AWS domain (SMART entry: SM00570), SET domain (Pfam ID: PF00856), and Post SET (SMART entry: SM00508) domain, but *MdSDG26.2* lost partial C-terminal sequence. And the expression levels of *MdSDG26.1* and *MdSDG26.2* both reached a peak at 24 hpi (Figure 1E,F). However, the expression of *MdSDG26.1* rose initially and then dropped, while *MdSDG26.2* was only highly expressed at 24 hpi.

3.2 | Two Transcript Variants of *MdSDG26* Have the Same Functions and Negatively Regulate *A. alternata* Resistance

To investigate whether the missed sequence in the *MdSDG26* C-terminus could affect the function or not, we performed subcellular localization, histone methyltransferases activity, and *A. alternata* inoculation experiment. Both *MdSDG26.1* and *MdSDG26.2* had the same subcellular localization to the nucleus (Figure 2A). Using transiently overexpressed plants of *MdSDG26.1* and *MdSDG26.2* (Figure 2B), we found that both *MdSDG26.1* and *MdSDG26.2* had the same histone methyltransferases activity of H3K4me3 and H3K36me3 (Figure 2C). In addition, both transiently overexpressed plants showed reduced resistance to *A. alternata* as compared to the wild-type

GL-3 (Figure 2D,E). The above results suggested that the C-terminal *MdSDG26* might not be necessary for the histone methyltransferases activity and *A. alternata* resistance.

To further verify the regulation of H3K4 and H3K36 trimethylation and *A. alternata* resistance by *MdSDG26*, we generated stable transgenic apple plants. We used the RNAi method to knockdown the expression of *MdSDG26* (including *MdSDG26.1* and *MdSDG26.2*, Figure 3A). Western blot analysis showed that the global level of H3K36me3 but not H3K4me3 was decreased (Figure 3B). After *A. alternata* was inoculated for about 10 days, *MdSDG26* knockdown plants were more resistant to *A. alternata* than GL-3 (Figure 3C).

Taken together, our results demonstrated that *MdSDG26* participates in the histone methylation of H3K4me3 and H3K36me3 and negatively regulates *A. alternata* resistance in apple plants.

3.3 | *MdSDG26* Mediates Genome-Wide Transcriptome Changes in Response to *A. alternata*

To understand the molecular basis for the effect of *MdSDG26* on apple *A. alternata* resistance, we performed RNA-sequencing to profile the genome-wide gene expression of GL-3 and *MdSDG26* RNAi plants under control (Mock inoculation) and disease treatment (*A. alternata* inoculated 24 h) conditions. We first analysed the differentially expressed genes (DEGs) regulated by *MdSDG26* under the control conditions by comparing the mock inoculation of GL-3 and *MdSDG26* RNAi plants (Supporting Information S1: Datasets 1, adjusted *p*-value < 0.01, and |fold change| > 2). Under control conditions, a total of 771 DEGs were identified with 504 upregulated genes and 267 downregulated genes in the *MdSDG26* RNAi plants compared with GL-3 plants (Figure 4A and Supporting Information S1: Dataset 1), respectively. After *A. alternata* inoculated for 24 h, 346 DEGs were upregulated, 306 DEGs were downregulated relative to Mock inoculation in GL-3 plants (Figure 4B and Supporting Information S1: Dataset 2), and 707 DEGs were upregulated and 1,599 DEGs were downregulated relative to Mock inoculation in *MdSDG26* RNAi plants (Figure 4C and Supporting Information S1: Dataset 3). Venn diagrams showed that 58 DEGs regulated by *MdSDG26* responded to *A. alternata* infection, and 397 genes were responsive to *A. alternata* infection in both GL-3 and *MdSDG26* RNAi plants (Figure 4D).

To clarify the pathway that is regulated by *MdSDG26* that confers *A. alternata* resistance in apples, we performed Gene Ontology (GO) terms analysis and Gene Regulatory Network (GRN) construction using the DEGs in *MdSDG26* RNAi plants in response to *A. alternata* infection. These DEGs with GO terms related to biological processes such as 'defense response (GO:0006952)', 'innate immune response (GO:0045087)', 'immune system process (GO:0002376)', and 'immune response (GO:0006955)' (Figure 4E and Supporting Information S1: Dataset 4). The hub of GRN was *MD01G1093700*, which is similar to *At4G35580* encoding NAC transcription factor-like 9 (NTL9, Figure 4F). Both results demonstrated that *MdSDG26*

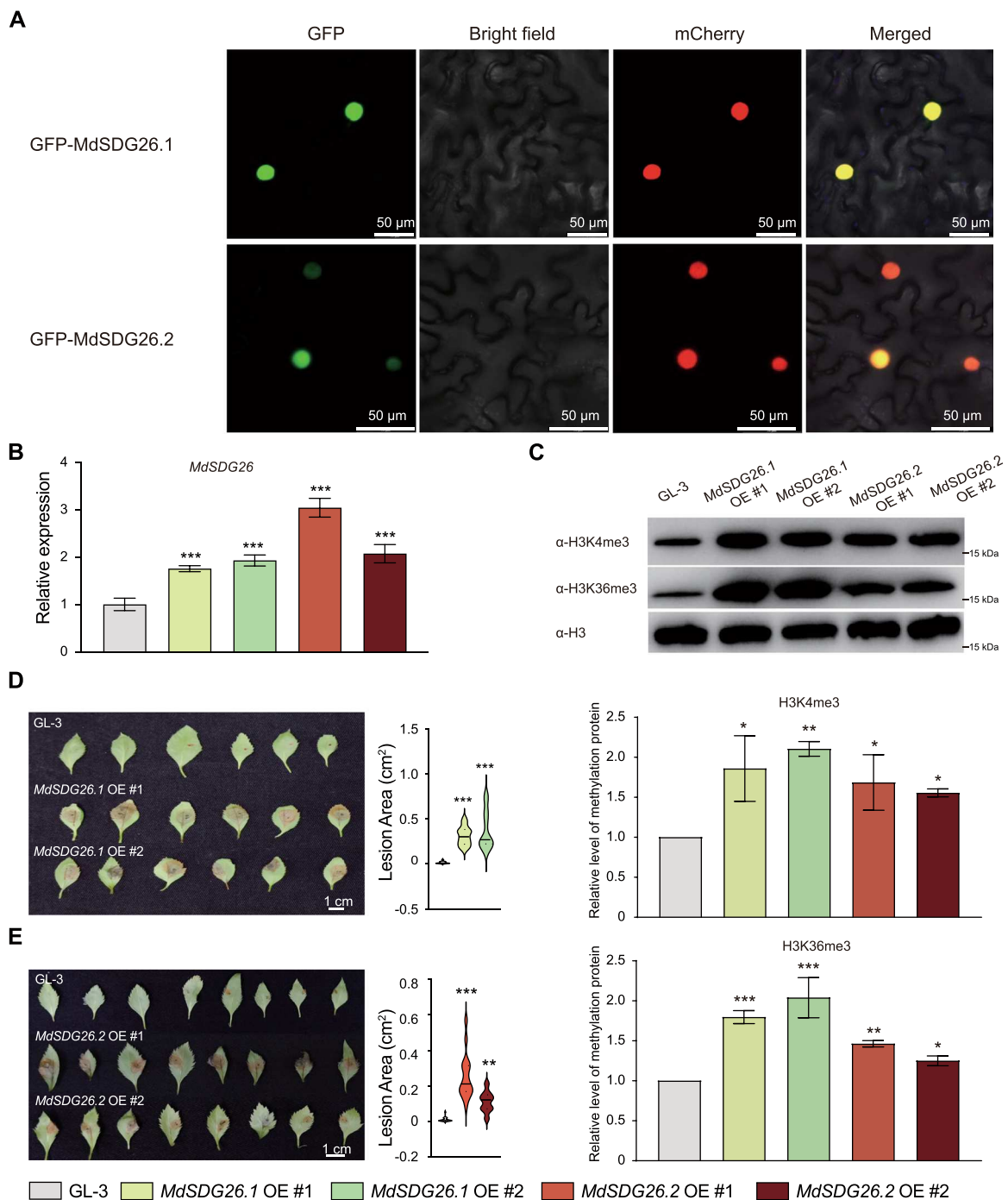


FIGURE 2 | Two *MdSDG26* variants have no functional difference. A, Subcellular localization of *MdSDG26.1* and *MdSDG26.2*. B, The transcription levels of *MdSDG26* in two independent 35S: *MdSDG26.1* transient overexpression parallel experiment and two independent 35S: *MdSDG26.2* transient overexpression parallel experiment. OE, overexpression. C, Western blot analysis and relative quantitative analysis of global H3K4me3 and H3K36me3 levels in GL-3, *MdSDG26.1* OE and *MdSDG26.2* OE plants. The Western blot data shown are representative of results consistently observed across three independent experimental replicates. D, E, Infection symptoms (left) and the lesion areas (right violin plots) of GL-3, *MdSDG26.1* OE (D) and *MdSDG26.2* OE (E) plants after *A. alternata* infected 5 days. Error bars indicate SD, $n = 3$ in (B and C) and $n = 40$ in (D and E). Statistical analyses were performed by one-way ANOVA (Tukey's test). Asterisks indicate significant differences between the transgenic lines and GL-3. Statistically significant differences are indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

indeed plays an important role in the regulation of *A. alternata* resistance.

To confirm the RNA-seq results, six disease-related genes defined based on data from *Arabidopsis* (Dixit, Gupta, and

Senthil-Kumar 2018; Hoang et al. 2021; Sugimoto et al. 2022; Nobori et al. 2023; Wang et al. 2023) were selected from DEGs for expression verification by RT-qPCR, and the expression of all genes could be verified (Figures 5 and S3). Among them, *MD01G1093700* (*MdNTL9*) and *MD01G1093700* (*MdNTL9-like*)

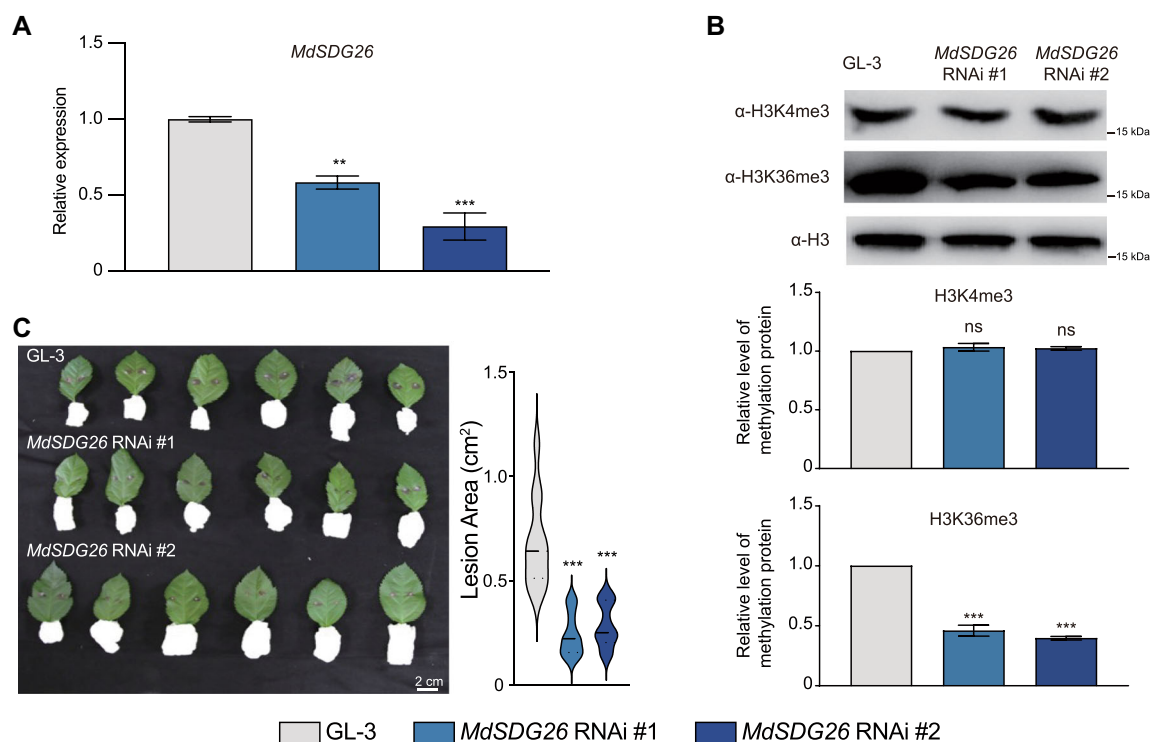


FIGURE 3 | Knockdown of the expression of *MdSDG26* negative modulates histone methylation and confers disease resistance. A, The transcription levels of *MdSDG26* in two independent *MdSDG26* RNAi lines. B, Western blot analysis and relative quantitative analysis of global H3K4me3 and H3K36me3 levels in GL-3 and *MdSDG26* RNAi plants. The Western blot data shown are representative of results consistently observed across three independent experiments. C, Infection symptoms (left) and the lesion areas (right violin plot) of GL-3 and two independent *MdSDG26* RNAi lines after *Alternaria alternata* infected 2 weeks. The experiment was conducted three times, yielding consistent trends. One representative result is presented. Error bars indicate SD, $n = 3$ in (A and B), and $n = 30$ in (C). Statistical analyses were performed by one-way ANOVA (Tukey's test). Asterisks indicate significant differences between the transgenic lines and GL-3, and statistically significant differences are indicated by ** $p < 0.01$, *** $p < 0.001$. ns, no significant difference. The results shown are representative of three independent experiments. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

were lower expressed in *MdSDG26* RNAi plants than GL-3 and could be repressed by *A. alternata* infection. *MD01G1093000* (*MdNAC050*) and *MD02G1128300* (*MdHK1*) were lower expressed in *MdSDG26* RNAi plants than GL-3 but not in response to *A. alternata* infection. The expression of *MD08G1104200* (*MdNB-ARC*) and *MD01G1074700* (*MdGBF3*) were higher in *MdSDG26* RNAi plants than GL-3 but no changes in response to *A. alternata* infection. Thus, *MdSDG26* could regulate the disease-related genes to regulate *A. alternata* resistance in apples.

3.4 | *MdSDG26* Regulates Histone Methylation Levels of Disease-Related Genes

Since knocking down *MdSDG26* expression affects the global level of H3K36 trimethylation, we hypothesized that *MdSDG26* mediates the H3K36me3 level at the chromatin of disease-related genes in response to *A. alternata* infection. Based on the GRN construction and the RNA-seq results, we selected *MdNTL9* for further research. *MdNTL9* is the most similar to *AtNTL9* according to the BLASTP results, and both proteins contain a conserved NAM domain (Figure 6A, Pfam ID: PF02365). We then examined the H3K36me3 and H3K4me3 levels of *MdNTL9* in response to *A. alternata* infection. ChIP-qPCR analysis showed that the H3K36me3 and H3K4me3 levels

of the *MdNTL9* promoter and exon regions were decreased after *A. alternata* infection. Specifically, the promoter and exon regions of *MdNTL9* had lower H3K36me3 and H3K4me3 levels in *MdSDG26* RNAi plants than in GL-3 whether under control or *A. alternata* infected conditions (Figure 6B). Since the identity of *MdNTL9* and *AtNTL9* was relatively low (46%), we examined the biological function of *MdNTL9* in *A. alternata* resistance. We transiently overexpressed *MdNTL9* in GL-3 leaves and inoculated *A. alternata* after being identified by RT-qPCR (Figure 6C). We found that *MdNTL9* played a negative role in *A. alternata* resistance (Figure 6D). Then, to investigate the link between *MdSDG26* at the *MdNTL9* chromatin, gene expression and their immune response functions, we transiently overexpressed *MdNTL9* in *MdSDG26* RNAi leaves and inoculated *A. alternata* after identification by RT-qPCR (Figure 6E). We found that *MdSDG26* RNAi/*MdNTL9* OE plants had a larger disease lesion size (Figure 6F). The above results demonstrated that *MdSDG26* regulates the histone H3K36 and H3K4 trimethylation levels at the chromatin of *MdNTL9* in response to *A. alternata* infection.

Besides *MdNTL9*, we also found another NAC transcription factor, *NAC81*, related to disease resistance which was reported previously (Delessert et al. 2005; Wang et al. 2009; Wang and Culver 2012). *MdNAC81* decreased expression in *MdSDG26* RNAi plants compared with in GL-3 under *A. alternata* infected

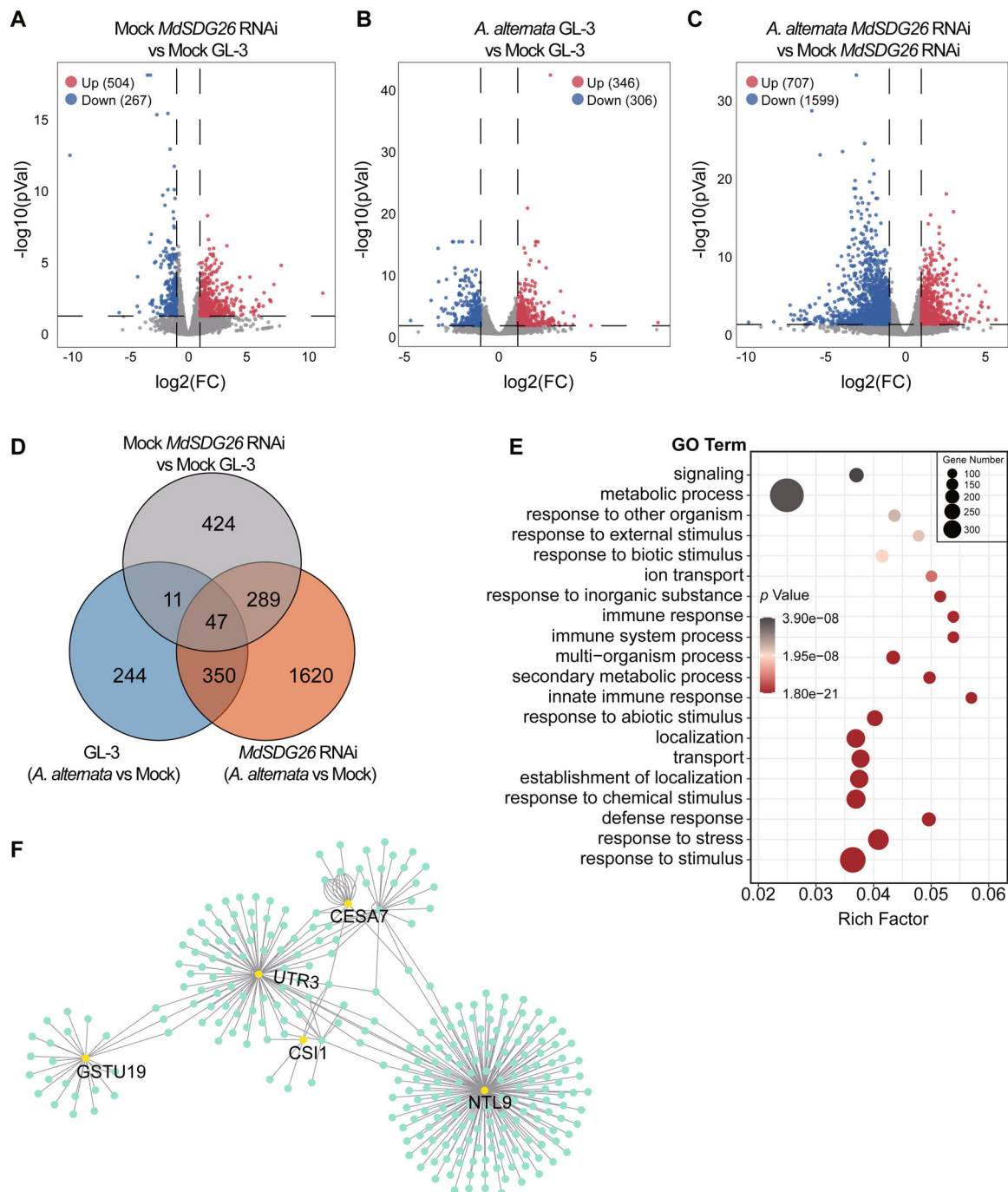


FIGURE 4 | *MdSDG26* regulates genome-wide transcriptome in response to *Alternaria alternata*. A, Volcanic diagram displayed the differentially expressed genes (DEGs) among *MdSDG26* RNAi plants and GL-3 under control conditions. B, Volcanic diagram displayed the DEGs between control and *A. alternata* treatment among GL-3 plants. C, Volcanic diagram displayed the DEGs between control and *A. alternata* treatment among *MdSDG26* RNAi plants. D, Venn diagrams presented the overlapped DEGs among three pairwise comparisons. E, Top 20 gene ontology (GO) terms in the biological process (BP) assigned from the DEGs between *MdSDG26* RNAi plants with GL-3 under control conditions. F, Gene regulatory network of the differentially expressed genes regulated by the interaction between *A. alternata* and genotype. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

conditions (Figure S4A). *MdNAC81* is most similar to *AtNAC81* according to the BLASTP results, and both proteins contain a conserved NAM domain (Pfam ID: PF02365), but *MdNAC81* had two unique internal repeat 1 domains (Figure S4B). ChIP-qPCR analysis demonstrated that the H3K4me3 and H3K36me3 levels at the promoter and exon regions of *MdNAC81* were decreased under control conditions. Furthermore, the promoter

and exon regions of *MdNAC81* exhibited lower H3K36me3 levels in *MdSDG26* RNAi plants compared to GL-3 plants, irrespective of whether the plants were under control conditions or *A. alternata* infected conditions (Figure S4C).

Based on the above results of this study, we propose a simple model elucidating the role of *MdSDG26* in the response to *A. alternata*

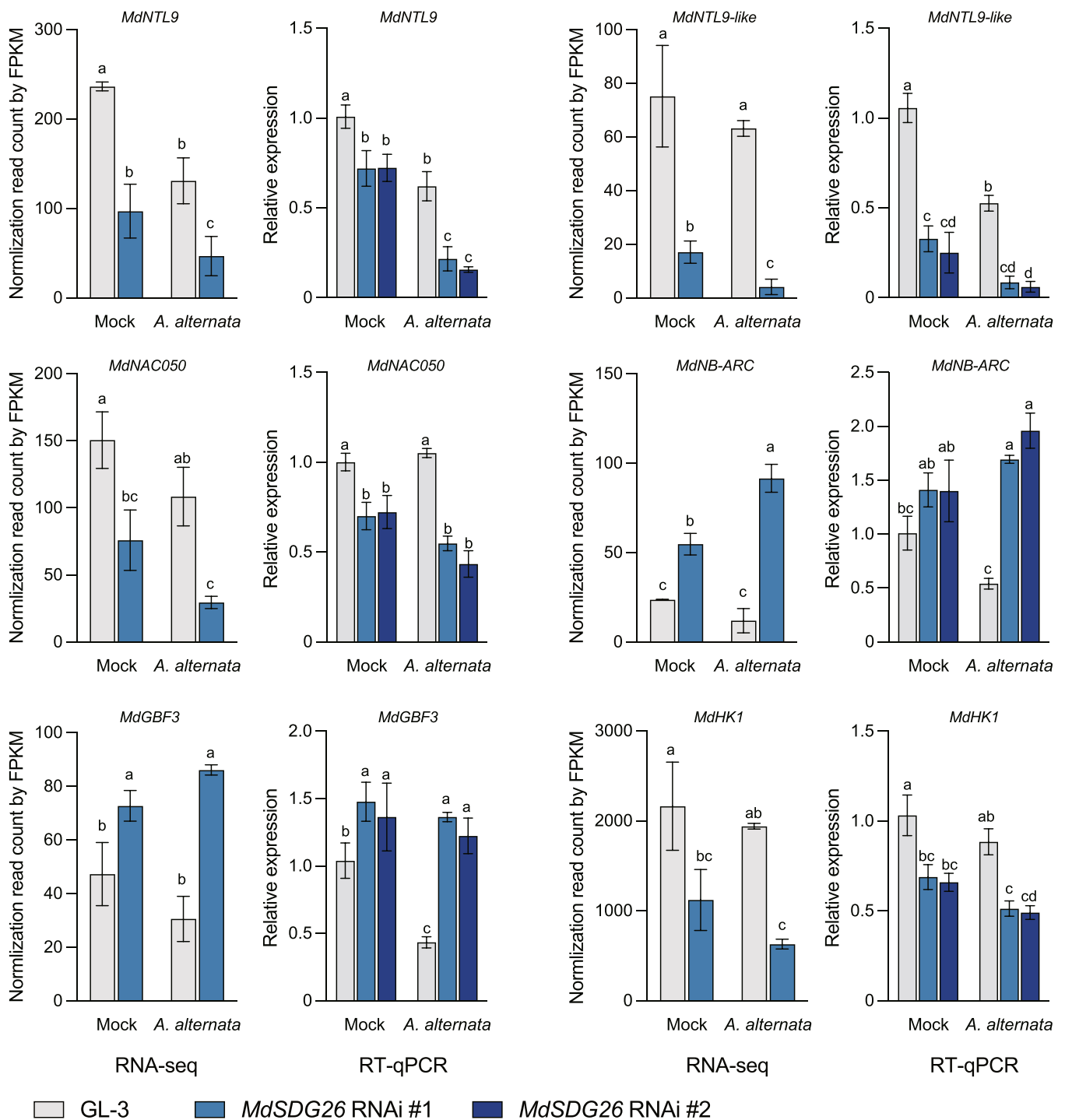


FIGURE 5 | Verification of RNA-seq data. Error bars indicate SD, $n = 3$. Statistical analyses were performed by one-way ANOVA (Tukey's test). The letters indicate significant differences ($p < 0.05$). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pe.15136)]

infection in apple (Figure 7). Upon inoculation with *A. alternata*, there is an induction in the expression of *MdSDG26*, a histone methyltransferase. This induction subsequently alters H3K36me3 levels at the *MdNAC81* locus, thereby enhancing *MdNAC81* expression. Additionally, *MdSDG26* enhances the expression of various other genes associated with disease response. Crucially, *MdSDG26* also regulates the levels of H3K4me3 and H3K36me3 at the *MdNNTL9* locus, a negative regulator of *A. alternata* resistance. This regulation results in increased expression of *MdNNTL9*, significantly contributing to the pathogenesis of apple leaf spot.

4 | Discussion

To deal with disease infection, plants have developed various immune systems, such as PTI and ETI (Pruitt et al. 2021). Previous studies have established that histone lysine methyltransferases (HLMTs) play important roles in disease resistance (Berr et al. 2010; Bvindi et al. 2022). In this study, we identified a SET-domain containing protein, *MdSDG26*, that negatively regulates *A. alternata* resistance by mediating histone H3K36 trimethylation level of disease-resistant genes.

The transposable elements in the genome often affect gene expression (Niu et al. 2022) or lead to novel transcripts (Chen et al. 2012; Sahebi et al. 2018). The quantity and alternative splicing of mRNA can be influenced by sequences produced from TEs. TEs inserted into introns are detected by the splicing machinery and recruited into RNA transcripts as exons in a process known as exonization. This process is based on TEs, which possess a multitude of splice-donor and splice-acceptor sites (Drongitis et al. 2019; Gebrie 2023). We found that *MdSDG26* has two transcript variants, *MdSDG26.1* and *MdSDG26.2*, which are caused by the transposable element insertion after the sixth exon of *MdSDG26*. Based on the protein alignment of *MdSDG26.1* and *MdSDG26.2* with *AtSDG26* in Arabidopsis, *MdSDG26.1* is the most similar transcript with *AtSDG26* in apple plants (Figure 1D). In the SDG group family, the functional domains are more important than the other regions. For example, the full-length SUVH3, which belongs to SDG group proteins, could bind the methylated DNA, but only the functional domains could also play the same function in vitro (Harris et al. 2018). Consistent with this report, in our study, both *MdSDG26.1* and *MdSDG26.2* belong to the class II SDG family, which contains the typical domains of the AWS domain, SET domain and Post-SET domain (Figure 1D). *MdSDG26.1* and *MdSDG26.2* played the same roles in regulating histone methylation, subcellular localization and *A. alternata* resistance (Figure 2), suggesting that the TE insertion in the intron/exon of *MdSDG26* does not affect its histone methyltransferase domain and *A. alternata* resistance, but results in two transcripts. However, the expression of *MdSDG26.1* and *MdSDG26.2* both reach a peak at 24 hpi, but the observed trends in expression volume are not entirely consistent (Figure 1E,F), which may be caused by the transposable element insertion, and further research is required to elucidate the precise regulatory mechanism.

In our study, the global levels of H3K4me3 and H3K36me3 were increased in *MdSDG26.1* and *MdSDG26.2* transiently overexpressed apple plants under normal conditions (Figure 2). However, we only observed reduced H3K36me3 level in *MdSDG26* RNAi plants under normal conditions (Figure 3), which might be due to the regulation by other HLMTs with H3K4me3 stabilization in *MdSDG26* RNAi plants under normal conditions. However, the level of H3K4me3 in the promoter and exon regions of *MdNLT9* and *MdNAC81* was found to be reduced in *MdSDG26* RNAi plants, which is similar to that observed in *sdg26* (Liu et al. 2016). This suggests that *MdSDG26* can regulate the level of

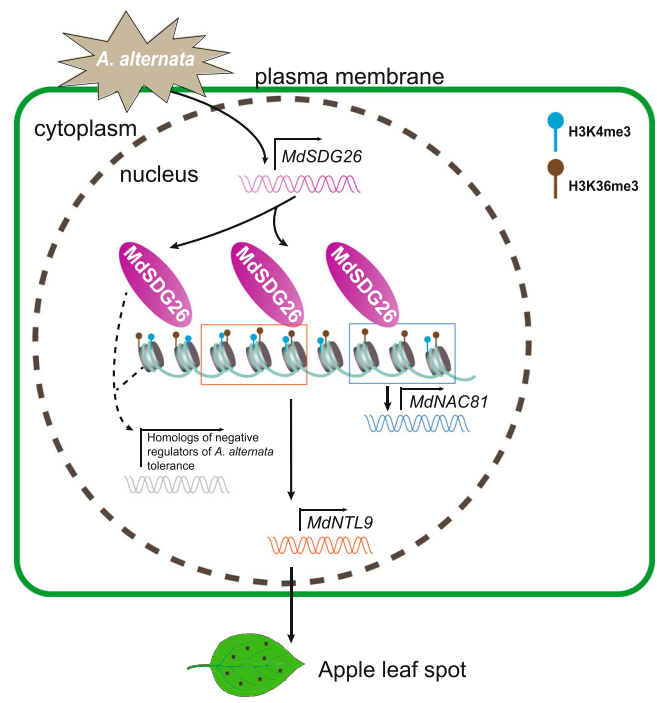


FIGURE 7 | A proposed model for the role of *MdSDG26* in response to *Alternaria alternata* infection in apple. Inoculation with *A. alternata* induces the expression of *MdSDG26*, a histone methyltransferase, which subsequently modulates the levels of H3K36me3 at the *MdNAC81* locus, thereby enhancing *MdNAC81* expression. Additionally, *MdSDG26* upregulates the expression of various other disease-related genes. Furthermore, a critical aspect is that *MdSDG26* regulates the levels of H3K4me3 and H3K36me3 at the *MdNLT9* locus, a negative regulator of *A. alternata* resistance. This regulation enhances the expression of *MdNLT9*, thereby playing a significant role in the pathogenesis of apple leaf spot. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pe.15136)]

H3K4me3 in specific regions. In Arabidopsis, *AtSDG26* interplays with *AtSDG8*, regulating transcription and plant flowering and development. *sdg8* but not *sdg26* mutant plants have global reduction of H3K36me3 and elevation of H3K36me1 (Liu et al. 2016). Although the *MdSDG26* sequence was more similar to the *AtSDG26* sequence, it also has a relatively high identity with *AtSDG8* between the conserved domains (Figure S2). In addition, the H3K36me3 pattern in *MdSDG26* RNAi apple plants was more similar to that in *sdg8* mutant plants in Arabidopsis (Figure 3B,

FIGURE 6 | *MdNLT9* as the potential target gene regulated by *MdSDG26* mediates *Alternaria alternata* resistance. A, Protein sequence alignment of *MdNLT9* and *AtNLT9*. At, *Arabidopsis thaliana*. Conserved NAM domains were indicated by blue squares. B, Relative enrichment of H3K4me3 and H3K36me3 at target loci in Mock or *A. alternata*-infected plants. The top panel showed the predicted genome structure of *MdNLT9*, exons and introns were indicated by boxes and lines, the grey box indicated UTR, the gene transcript direction was marked as the arrows, and the primer used for ChIP-qPCR was indicated at the top of structure diagram. Error bars indicate SD, $n = 3$. Statistical analyses were performed by one-way ANOVA (Tukey's test). The letters indicate significant differences ($p < 0.05$). C, The transcription levels of *MdNLT9* in 35S: *MdNLT9* transient overexpression lines. D, Infection symptoms (left) and the lesion areas (right violin plot) of GL-3, *MdNLT9* OE plants after *A. alternata* inoculated for 5 days. E, The transcription levels of *MdNLT9* in 35S: *MdNLT9* transient overexpression in *MdSDG26* RNAi lines. F, Infection symptoms (left) and the lesion areas (right violin plot) of *MdSDG26* RNAi control lines and *MdSDG26* RNAi/*MdNLT9* OE plants after *A. alternata* inoculated 5 days. The experiments were conducted twice, yielding consistent trends. One representative result is presented. Error bars indicate SD, $n = 3$ in (C and E), and $n = 30$ in (D and F). Statistical analyses were performed by Student's t test. The asterisks indicate significant differences, and statistically significant differences are indicated by ** $p < 0.01$, *** $p < 0.001$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/pe.15136)]

Liu et al. 2016). Given that SDG8 regulates histone methylation and regulates pep1-, flg22- and effector-triggered immunity as well as systemic acquired resistance in Arabidopsis (Lee et al. 2016), while the homology genes of SDG8 (SDG33 and SDG34) in tomato negatively regulated the disease resistance (Bvindi et al. 2022), and MdSDG26 negatively regulated disease resistance (Figures 2 and 3), suggests that although these HLMTs are orthologous with similar biochemical activities but exhibit different physiological functions across plant species. It is of greater importance to identify the components that recruit these HLMTs to specific target loci in apples, as this will help to elucidate the specific roles and regulatory mechanisms of these HLMTs, which is crucial for the potential utilization of these genes.

The responses of plants to pathogens and environmental stress involve the rapid reprogramming of gene transcription, which occurs in the context of chromatin, such as histone modification and chromatin accessibility (Bvindi et al. 2022). The ChIP data indicated that MdSDG26 regulates the induced HLM and subsequent reprogramming of gene expression. A network analysis of genes regulated by *A. alternata* in an MdSDG26-dependent manner revealed that one of the most regulated genes of this network was MdNLT9, an orthologue of Arabidopsis NTL9. Previous reports have shown that NTL9 functions as a suppressor of disease resistance by influencing salicylic acid production, basal resistance, effector-triggered immunity and stomatal immunity (Wang et al. 2009; Guo et al. 2017; Marín-Ponce et al. 2023). In line with these findings, MdNLT9 in apple trees was observed to have a negative impact on resistance to *A. alternata* (Figure 6C,D). Additionally, in MdSDG26 RNAi plants, expression of MdNLT9 and MdNLT9-like, the two homologous genes of NTL9 in Arabidopsis, was lower than that in GL-3 after *A. alternata* infection (Figure 5). Histone H3K4 and H3K36 trimethylation are typically responsible for active transcription (Berger 2007). In the promoter and exon regions of MdNLT9, the enrichment of H3K36me3 and H3K4me3 was lower in MdSDG26 RNAi plants than in GL-3 in both control and *A. alternata* infected condition, and the enrichment level of H3K36me3 and H3K4me3 in the promoter and exon regions of MdNLT9 was repressed by *A. alternata* infection, consistent with the reduced expression of MdNLT9 in MdSDG26 RNAi plants. Moreover, our findings indicate that overexpressed MdNLT9 in MdSDG26 RNAi plants could recover susceptibility (Figure 6E,F). These combined outcomes suggest that MdNLT9 contributes to the increased resistance of MdSDG26 RNAi plants to *A. alternata* infection, and it is a keystone gene that is regulated by MdSDG26 in *A. alternata* resistance.

NAC81 overexpressed in Arabidopsis displays a higher susceptibility to the soil-borne fungal pathogen *F. oxysporum* (Delessert et al. 2005; Wang et al. 2009; Wang and Culver 2012). In our study, we observed that there were consistent reductions in H3K36me3 levels in both the promoter and exon regions of MdNAC81 in MdSDG26 RNAi plants, both in the absence and presence of *A. alternata* infection. However, reductions in H3K4me3 levels in both the promoter and exon regions were only observed in MdSDG26 RNAi plants under control conditions (Figure S4C), suggesting that the expression and chromatin of MdNAC81 was also affected by MdSDG26 and may

play an important role in *A. alternata* resistance (Figure S4). The regulation of other disease-related genes, such as NB-ARC domain-containing proteins belonging to R proteins that contribute to disease resistance (Wang et al. 2023), is also of interest. GBF3 belongs to the basic leucine zipper (bZIP) transcription factor family and may have a role in combined stress tolerance in Arabidopsis, including disease resistance (Dixit, Gupta, and Senthil-Kumar 2018). HK1 belongs to the family of histidine kinases (HKs), which have been demonstrated to mediate plant responses to pathogens (Hoang et al. 2021). NAC050 belongs to the NAC-domain-contained transcription factor family and may participate in plant immunity (Nobori et al. 2023). These integrated results and RNA-seq results demonstrate the broader impacts of MdSDG26 in regulating disease-related gene expression. However, the function of MdGBF3, MdNB-ARC, MdHK1 and MdNAC50 in apple *A. alternata* resistance remains unclear. Their contributions to disease-related gene expression have been defined based on data from Arabidopsis, but further research is needed to determine their biological functions.

In conclusion, we identified a SET-domain containing protein MdSDG26, which regulates the histone H3K4 and H3K36 trimethylation and negatively regulates *A. alternata* resistance. MdNLT9 seems to be the major contributor among the disease-related genes regulated by MdSDG26 through H3K36me3 and H3K4me3, which plays a negative role in *A. alternata* resistance (Figure 7). Our study provides information about the role of MdSDG26 in response to *A. alternata* infection, which will be useful for breeding disease-resistant apple plants by genetic engineering in the future.

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Data Availability Statement

RNA-seq data in this study have been submitted to NCBI under the project PRJNA1026094. Sequence data used in this article can be found in GenBank or GDR (Genome Database for Rosaceae) under the following accession numbers: OR651352 (*MdSDG26.1*), OR651353 (*MdSDG26.2*), MD01G1092200 (*MdNLT9*), MD01G1093700 (*MdNLT9-like*), MD01G1093000 (*MdNAC050*), MD08G1104200 (*MdNB-ARC*), MD01G1074700 (*MdGBF3*), MD02G1128300 (*MdHK1*), MD01G1092300 (*MdNAC81*).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.